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Eulerian SIR model for Asymmetric Temporal Commuting Networks: Case Study of Chinese Baidu Mobility Data

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Abstract. *SIR models for metapopulations are widely used to study infectious disease spread but often assume static networks or fixed flows. Temporal mobility networks, however, exhibit dynamic variations that impact epidemic propagation. This paper hypothesizes that traditional epidemiological models can be adapted to incorporate dynamic inflows and outflows while preserving population conservation. We adapt a metapopulation SIR model to temporal mobility networks using real mobility data from China (January–February 2020) at the onset of COVID-19. The study examines whether these variations improve epidemic spread modeling. Results suggest that this adaptation may reveal dynamic dissemination patterns that could be overlooked by conventional approaches, highlighting the importance of considering temporal mobility variations in epidemic modeling.*

1. Introduction

A complex network is a graph composed of vertices, representing entities that comprise the complex system under study, and edges, which capture the interactions between them. Examples include mobility networks, the Internet, and social networks [Barabási and Pósfai 2016].

An important characteristic of real-world networks is their evolution over time [Kim and Anderson 2012, Masuda et al. 2021]. There are various ways to represent temporal networks, such as aggregated networks and event, transmission, and reachability graphs [Sano 2021]. Common metrics, such as closeness (c_i) and betweenness (b_i), must be adapted depending on the type of representation, as the notions of connectivity and shortest path are, in these cases, tied to time.

Mobility networks are characterized by nodes representing locations and links denoting the flow of people or vehicles between nodes within a given time window [Lamosa et al. 2021]. In such networks, some datasets provide static data with one or more layers [Cavararo 2017, Governo do Estado de São Paulo 2019], while others include both layers and a temporal component [Gallotti and Barthelemy 2015]. Epidemiological processes, for instance, can be simulated with greater detail as the network representation becomes more comprehensive.

This project addresses a gap in the literature regarding epidemiological models for temporal mobility networks that account for both inflow and outflow dynamics.

While many studies rely on static graphs [Lipshtat et al. 2021, Anupriya et al. 2022], mobility data such as the Baidu Mobility Data [China Data Lab 2020] reveal temporal fluctuations in edges flows. We propose adapting SIR metapopulation models [Kermack and McKendrick 1927, Harko 2014] to incorporate dynamic inflows and outflows while ensuring the total number of individuals remains constant, even with mobility rates.

The main contribution of this project lies in the explicit modeling of temporal mobility through the use of inflow and outflow tensors, which capture time-varying movements between nodes. Unlike prior studies that rely on static networks or average mobility matrices, our model accounts for daily mobility fluctuations and ensures population conservation at the node level. This approach enhances fidelity and allows the model to reflect abrupt mobility shifts.

2. Materials and methods

2.1. Network analysis

Mathematically, a network is represented as a graph $G(V, E)$, where V denotes the set of nodes and E the set of edges. The total number of nodes, K, corresponds to the cardinality of V, while the total number of edges, L, is the cardinality of E. Networks can also be weighted, with each edge (i, j) assigned a weight w_{ij} representing the strength or intensity of the connection. The adjacency matrix A is a fundamental representation, where $A_{ij} = 1$ if there is a link from i to j, and $A_{ij} = 0$ otherwise [Barabási and Pósfai 2016].

A path is a sequence of connections between two nodes, and the shortest path is the one that requires the fewest connections, or, in weighted networks, the one with the least cost, based on the weights of the connections. The distance between two nodes, often called the geodesic distance d_{ij}, is the length of this shortest path, which offers insight into how quickly information or resources could travel between them. For network level metrics, we calculate the average path length $\langle d \rangle$, to assess how connected the network is, giving an overall view of its efficiency or cohesiveness. The diameter l_{max} of the network is the longest of all the shortest paths between any pair of nodes, representing the maximum distance that can be found in the network.

In weighted networks, the interpretation of distance is influenced by what the weights represent. If the weights reflect physical distance, the distance between nodes will directly correspond to these values. However, when weights represent factors like traffic or flow, the distance can be defined as the inverse of these weights, so that stronger connections or higher flows lead to shorter distances, highlighting the more efficient paths through the network. This is often expressed as l_{ij}, where the distance is calculated as the sum of the inverses of the weights along the shortest path.

Centrality metrics further deepen our understanding by indicating the relative importance of nodes within a network [Newman 2018]. Closeness centrality c_i tells us how close a node is to all other nodes, with more central nodes being those that are closer to others. The closeness centrality can be expressed as:

$$c_i = \frac{K - 1}{\sum_{j \neq i} d_{ij}}. \tag{1}$$

Betweenness centrality b_i identifies nodes that serve as bridges between different parts of the network, as these nodes connect otherwise separate regions. If removed, these nodes would disrupt the flow of information or resources, emphasizing their critical role in maintaining the network's structure. Betweenness centrality is given by:

$$b_i = \sum_{j \neq k \neq i} \frac{\sigma_{jk}(i)}{\sigma_{jk}}, \tag{2}$$

where σ_{jk} is the total number of shortest paths from node j to node k, and $\sigma_{jk}(i)$ is the number of those paths that pass through node i.

These metrics help to reveal the underlying structure of the network, shedding light on which nodes are most critical for maintaining connectivity or facilitating movement through the system. Understanding the centrality of nodes enables us to identify key players that shape the network's efficiency and resilience.

As we analyze networks, it is also important to consider their temporal aspects. Temporal networks extend the concept of static networks by incorporating the timing of interactions, enabling a more accurate representation of systems where the sequence and duration of connections matter [Holme and Saramäki 2013]. For example, in human mobility networks, the flow of people between locations varies over time, and capturing these dynamics is crucial for modeling phenomena like disease spread or information diffusion.

2.2. Compartmental models

Compartmental models, such as the SIR model, are widely used in epidemiology to simulate the spread of infectious diseases within populations [Keeling and Rohani 2008]. The SIR model consists of three compartments:

- **Susceptible (S)**: Individuals who are susceptible to disease.
- **Infectious (I)**: Individuals who have been infected and can transmit the disease.
- **Recovered (R)**: Individuals who have recovered and are no longer infectious.

The model for a population with N individuals is governed by the following system of ordinary differential equations:

$$\begin{cases} \dot{S} = -\frac{\beta SI}{N}, \\ \dot{I} = \frac{\beta SI}{N} - \gamma I, \\ \dot{R} = \gamma I, \end{cases} \tag{3}$$

where β is the transmission rate, measured in units of time^{-1}, representing the probability of an infected individual transmitting the disease to a susceptible individual per unit of time. The recovery rate γ, also in units of time^{-1}, represents the rate at which infected individuals recover or are removed from the infectious compartment. The total population $N = S + I + R$ remains constant over time, as shown by the conservation equation:

$$\dot{S} + \dot{I} + \dot{R} = 0. \tag{4}$$

The behavior of the SIR model is determined by the relationship between β and γ. If $\beta > \gamma$, the disease spreads through the population, leading to an epidemic. Conversely, if $\beta \leq \gamma$, the disease dies out quickly, as individuals recover faster than the disease can spread.

2.3. Metapopulations and host movement models

Metapopulations describe groups of populations of the same species occupying geographically separated areas and interacting at some level. This concept has been widely applied in ecology and epidemiology to understand how species or diseases respond to fragmented environments.

The integration of network and epidemiological models with metapopulation frameworks offers a more comprehensive perspective on understanding the spread of infectious diseases in complex systems. In this context, metapopulations can be seen as nodes within a complex network of spatial patches, where the links between them represent human mobility and facilitate disease transmission across patches.

Host movement models, such as Eulerian and Lagrangian approaches, are essential for simulating host movement within metapopulation networks. In our study, we adopt the Eulerian approach, which does not track the individual behavior of hosts, making it more suitable for describing large-scale migration [Citron et al. 2021]. To understand the geographical diffusion of diseases at varying spatial scales due to human mobility, epidemic modeling employs reaction-diffusion dynamics in metapopulations. Specifically, we use the Eulerian model to describe the movement of hosts between metapopulations, represented by the transport operator Ω_i [Barrat et al. 2008]:

$$\dot{N}_i = \Omega_i(\{N\}) = \sum_{j}^{K} f_{ji}\{N_j\} - f_{ij}\{N_i\}, \tag{5}$$

where N_i is the number of hosts at location i, among K subpopulations. By combining the SIR model with the Eulerian approach, we generate a set of $3K$ equations for the subpopulations:

$$\begin{cases} \dot{S}_i = -\frac{\beta_i S_i I_i}{N_i} + \Omega_i(\{S\}) = -\frac{\beta_i S_i I_i}{N_i} + \sum_j^K f_{ji}\{S_j\} - f_{ij}\{S_i\}, \\ \dot{I}_i = \frac{\beta_i S_i I_i}{N_i} - \gamma I_i + \Omega_i(\{I\}) = \frac{\beta_i S_i I_i}{N_i} - \gamma I_i + \sum_j^K f_{ji}\{I_j\} - f_{ij}\{I_i\}, \\ \dot{R}_i = \gamma I_i + \Omega_i(\{R\}) = \gamma I_i + \sum_j^K f_{ji}\{R_j\} - f_{ij}\{R_i\}. \end{cases} \tag{6}$$

2.4. Related works

Building on work by [Keeling and Eames 2005] and [Citron et al. 2021], which explore network structure and disease dynamics, we extend their efforts by incorporating temporal mobility networks and real-world data from Baidu Mobility [China Data Lab 2020]. Related studies, such as the GLEAM project [Vespignani et al. 2023] and research by [Karsai et al. 2011] and [Balcan et al. 2009], emphasize the role of network topology and temporal dynamics in disease spread. Our approach integrates dynamic mobility flows into the SIR framework, offering a more comprehensive tool for real-world analysis.

Additionally, while [Poletto et al. 2016] incorporated heterogeneous population interactions, our framework explicitly models temporal mobility dynamics.

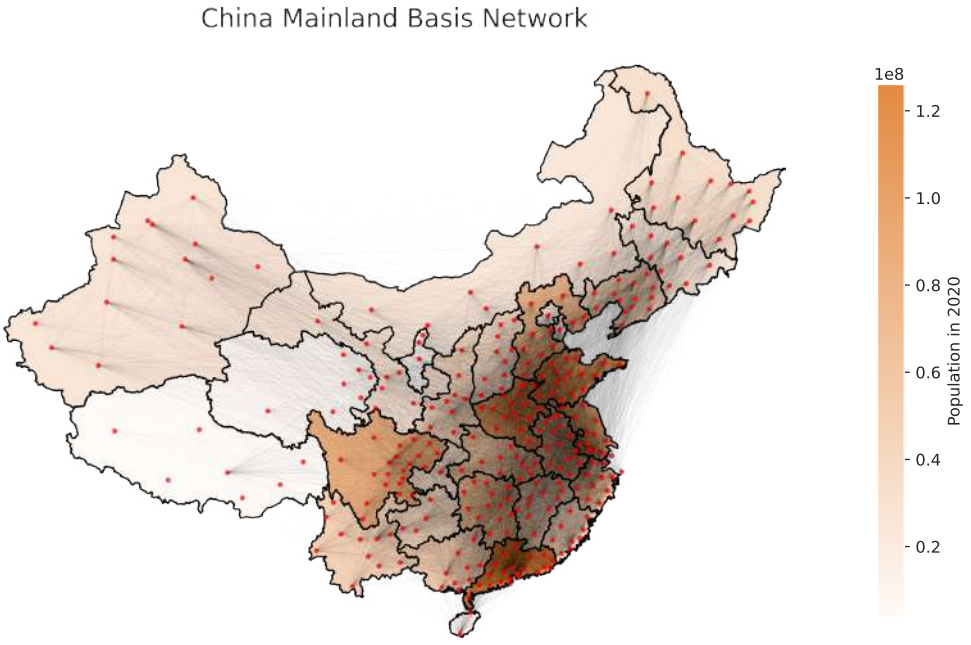
[Citron et al. 2021] applied SIR and Ross-Macdonald models using fixed mobility rates; we extend this by incorporating temporal variations. Adapting the transport operator framework from [Barrat et al. 2008], we include time-varying connectivity, complementing the GLEAM project's focus on multiscale mobility.

2.5. Mobility and geographical data

We use Baidu Mobility Data [China Data Lab 2020], collected through Baidu applications and services — the data are derived from Baidu's location-based services (LBS), which utilize a combination of GPS, IP addresses, Wi-Fi, and mobile app signals to estimate human mobility patterns across Chinese regions — covering January to February 2020. This period includes key events such as the Chunyun travel season (starting January 10), the Wuhan travel ban (January 23), the Lunar New Year (January 25), and the end of the Spring Festival (February 18).

Chunyun represents the world's largest annual migration, with over three billion trips recorded in 2020 [Wong 2020]. However, the onset of COVID-19 during this period led to strict travel restrictions, including the Wuhan lockdown, which suspended public transport and restricted movement to contain the outbreak [Tian 2020].

Baidu Mobility data is structured as daily inflow and outflow matrices for 369 Chinese regions. Each inflow matrix represents the rate of people traveling from city i to city j relative to total arrivals in j, while outflow matrices describe departures from j to other regions. Each node has up to 100 incoming and 100 outgoing connections. The structure of these flows is visualized in Figure 1, which shows the mobility network of mainland China on January 1, 2020, based on inflow data.



(a)

Figure 1. Mobility network of mainland China, January 1, 2020 (inflow).

Preprocessing by [Freitas et al. 2022] removed regions lacking geographical

codes, yielding daily flows between 303 regions over 60 days. The data is organized into tensors w_{ji}^{in} and w_{ij}^{out}, representing population movements at implicit time t, with dimensions days $\times$ nodes $\times$ nodes.

Major mobility hubs such as Beijing, Shanghai, and Wuhan play key economic and social roles. During COVID-19, Wuhan's lockdown significantly altered mobility patterns, concentrating over 20% of its outflow within Hubei province [Freitas et al. 2022]. Additionally, neighboring cities experienced a temporary surge in movement as people sought to exit before restrictions took effect [Pang et al. 2022].

Analyzing these mobility shifts is essential for understanding virus spread, as mobility fluctuations correspond with epidemic dynamics. Network metrics, including weighted betweenness (b_w), weighted closeness (c_w), and network diameter ($l_{\max}$), reveal substantial changes during the Wuhan lockdown [Tan 2021].

Figure 2 shows the temporal evolution of network metrics, highlighting significant changes in mobility patterns between January 23 and 25, corresponding to the Wuhan travel ban. Figure 3 displays a heatmap of Frobenius norms of network differences, where smaller values indicate similar mobility patterns, and larger values reflect disruptions. These variations are key for epidemic modeling, as they impact regional interactions and disease spread.

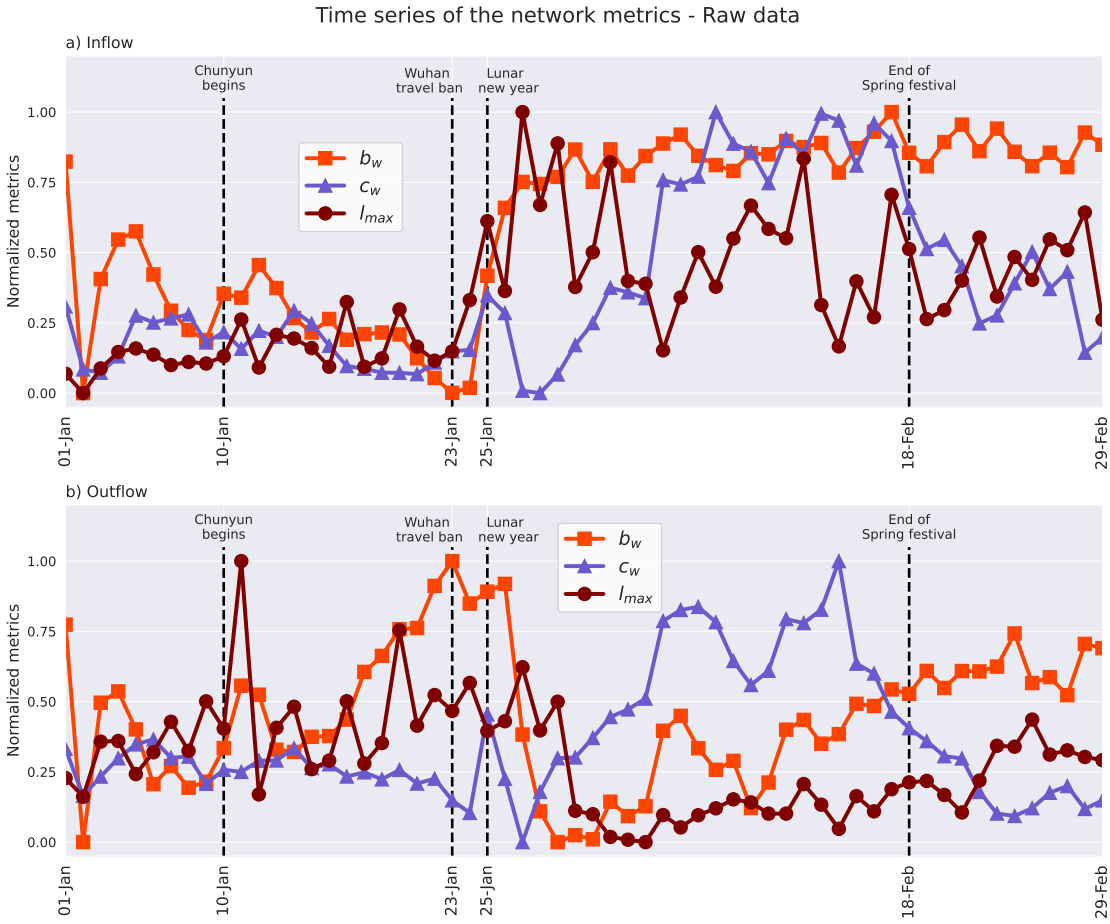


Figure 2. Temporal evolution of the network metrics during the analyzed period.

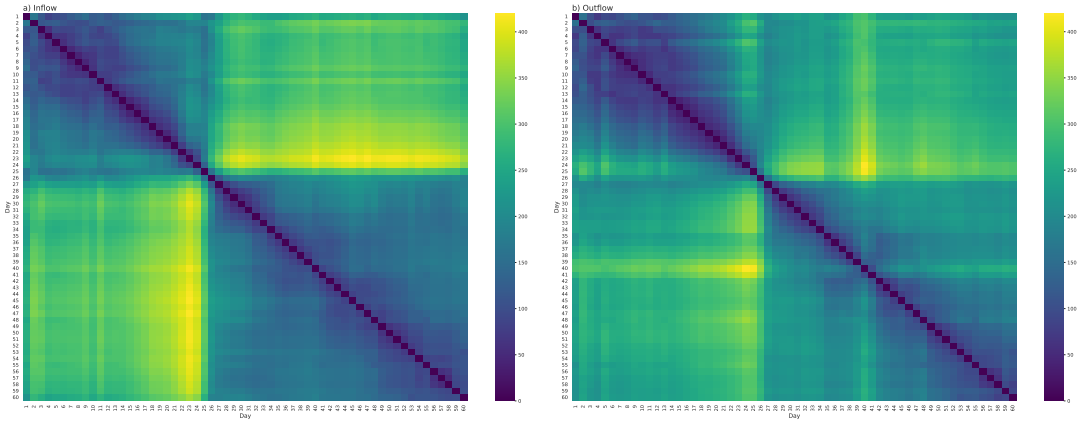


Figure 3. Heatmap of Frobenius norms between pairs of daily flows. Low norms indicate similar patterns pre-January 25, while high norms reflect changes after the restrictions.

For calibration, we scraped *City Population* [Brinkhoff 2024] to obtain 2020 population data, associating them with network nodes. Using geospatial bases, we structured a *GeoDataFrame* with node coordinates, polygons, area, perimeter, and aggregated provincial populations.

To maintain constant node populations in simulations, inflow and outflow adjustments were implemented iteratively to redistribute flows and minimize population imbalances. This ensures that the sum of inflows and outflows remains approximately constant by reallocating individuals based on each node's capacity, defined as a percentage of its initial population. The redistribution process is crucial to prevent artificial population variations over time, which could affect the consistency of the epidemiological model and the numerical integration of differential equations. A limiting factor on flows is also applied, capping the maximum percentage of travelers per node to ensure model stability and prevent unrealistic fluctuations that could distort epidemiological outcomes. The flow adjustment algorithm, detailed in Algorithm 1, captures this iterative process.

2.6. Eulerian SIR model with input and output flows

Understanding these dynamic mobility patterns is essential for accurately modeling disease transmission. To incorporate these variations into epidemic simulations, we adapt the Eulerian model and the Eulerian SIR model, allowing for a more flexible and realistic representation of host movement across metapopulation networks. Instead of using a single travel rate matrix, as in [Citron et al. 2021], we employ temporal inflow and outflow tensors.

The classical Eulerian model relies on a single matrix (f_{ij}) that describes host migration rates between subpopulations. However, this matrix imposes a static structure on flows, limiting its ability to represent temporal changes and the complexity of real networks. To overcome this limitation, we replace the travel rate matrix with two tensors: an inflow tensor (w_{ji}^{in}) and an outflow tensor (w_{ij}^{out}), both varying over time and across different network configurations.

These flow tensors describe the number of hosts arriving at node i from node j and those departing from i to j, respectively, at a given implicit time t. Consequently,

Algorithm 1 Population-Constrained Flow Adjustment

Require:
$\mathbf{P} \in \mathbb{R}^N$ ▷ Node populations
$\mathbf{F}^{in}, \mathbf{F}^{out} \in \mathbb{R}^{D \times N \times N}$ ▷ Flow tensors
$\alpha \in (0,1), \epsilon > 0, I_{max}$ ▷ Constraints
Ensure:
Adjusted flows $\mathbf{F}_*^{in}, \mathbf{F}_*^{out}$ satisfying $||\mathbf{F}_*^{out}||_1 \leq \alpha \mathbf{P}$
Initialize $\mathbf{T}_{max} \leftarrow \alpha \mathbf{P}$, $\mathbf{F}_*^{in} \leftarrow \mathbf{F}^{in}$, $\mathbf{F}_*^{out} \leftarrow \mathbf{F}^{out}$
for each day $d \in [1, D]$ **do**
  for $k \leftarrow 1$ **to** I_{max} **do**
    Compute imbalances $\boldsymbol{\delta} \leftarrow \sum_j (\mathbf{F}_*^{in})_{d,\cdot,j} - \sum_j (\mathbf{F}_*^{out})_{d,j,\cdot}$
    if $\|\boldsymbol{\delta}\|_\infty < \epsilon$ **then break** ▷ Converged
    end if
    for each node n **do**
      Scale inflows or outflows proportionally to δ_n ▷ Balance flows
    end for
  end for
  for each node n **do**
    Enforce $\|\mathbf{F}_*^{out}\|_1 \leq T_{max,n}$ ▷ Apply traveler cap
  end for
  Synchronize $\mathbf{F}_*^{in} \leftarrow (\mathbf{F}_*^{out})^\top$ ▷ Maintain reciprocity
end for

the transport operator Ω_i, which computes the flow balance for each subpopulation, is redefined to account for these temporal variations:

$$\dot{N}_i = \Omega_i(\{N\}) = \sum_j^K w_{ji}^{in}\{N_j\} - w_{ij}^{out}\{N_i\}, \tag{7}$$

where N_i represents the number of hosts in location i at time t (implicit), and the balance between inflows and outflows is continuously updated. This enables a dynamic and accurate representation of host redistribution among subpopulations while preserving the total number of individuals over time:

$$N = \sum_{i=1}^K N_i. \tag{8}$$

With this adaptation, the Eulerian SIR model for epidemic dynamics between subpopulations is modified as follows:

$$\begin{cases} \dot{S}_i = -\frac{\beta_i S_i I_i}{N_i} + \sum_j^K w_{ji}^{in}\{S_j\} - w_{ij}^{out}\{S_i\}, \\ \dot{I}_i = \frac{\beta_i S_i I_i}{N_i} - \gamma I_i + \sum_j^K w_{ji}^{in}\{I_j\} - w_{ij}^{out}\{I_i\}, \\ \dot{R}_i = \gamma I_i + \sum_j^K w_{ji}^{in}\{R_j\} - w_{ij}^{out}\{R_i\}. \end{cases} \tag{9}$$

Thus, each compartment (S_i, I_i, R_i) is updated based on temporal inflows and outflows, incorporating interactions between subpopulations and mobility dynamics. The use of flow tensors allows for a more accurate modeling of how epidemics spread in complex mobility scenarios, capturing both temporal and spatial heterogeneity in the network.

3. Results

The flow tensors, after being processed as described in Algorithm 1, were adjusted to limit mobility to 10% of each node's population. Since the Baidu mobility data provides mobility rates rather than absolute values, we cross-referenced these rates with the actual population of the municipalities. To simulate a more realistic scenario, a maximum of 10% of each municipality's population was allowed to move. With $\beta_i = 0.9$, $\gamma = 0.1$, and 1 infected individual per node, the RK45 [Dormand and Prince 1980] numerical method was used with a step size of $h = 10^{-3}$ s. Temporal integration was performed for each region of mainland China, as shown in Figures 4 and 5. Figure 4 displays the temporal evolution of Equation (7), while Figure 5 illustrates the dynamics of the system in Equation (9). The color bars illustrate all 303 regions. In both Figures, the y-axis represents the total number of individuals normalized by subpopulation or within each compartment of each subpopulation.

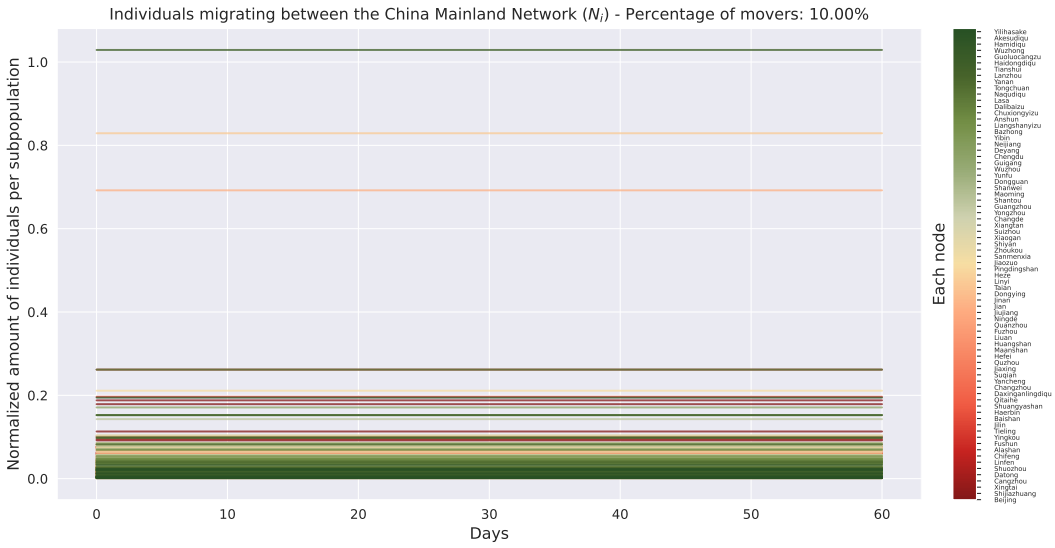


Figure 4. Time integration of Equation (7) **for each region of mainland China.**

The regions with the highest infection peaks were *Haidongdiqu*, *Laiwu*, and *Jian*. It is noteworthy that between January 20 and 25, a period marked by the Wuhan travel restrictions (January 23) and the Lunar New Year (January 25), the infection curves in most regions showed a sharp increase. This surge coincided with the intense population movement during the Spring Festival, highlighting how large-scale mobility events can accelerate disease spread. The Wuhan lockdown, although necessary, also led to a spillover effect, with unexpected increases in mobility in neighboring regions. This underscores the complexity of disease propagation dynamics in scenarios involving travel restrictions and massive population flows.

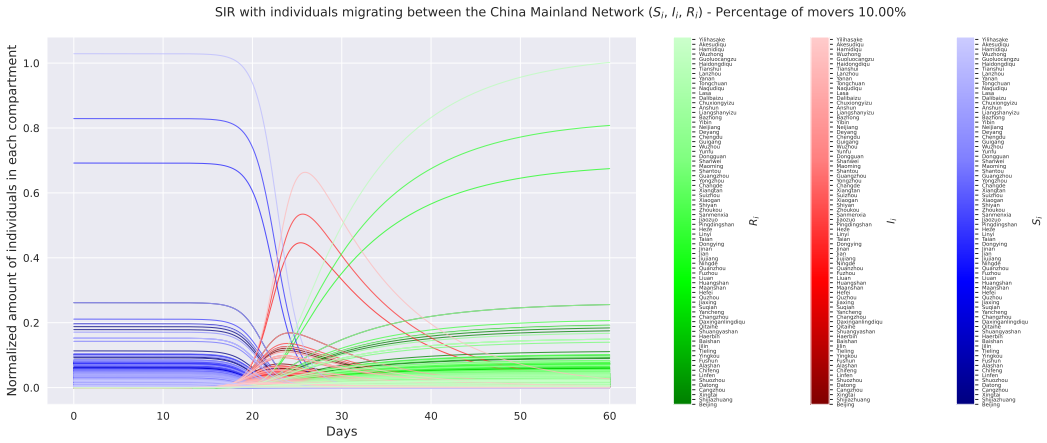


Figure 5. Time integration of the system of Equation (9) for each region of mainland China.

The temporal network analysis (Figures 2 and 3) reveals three evidence-based relationships with the SIR epidemic dynamics (Figures 4 and 5): (1) The abrupt changes in network metrics after January 23 (Figure 2) coincide with accelerated infection growth in multiple regions, demonstrating how mobility pattern shifts affected transmission rates. (2) The increased flow dissimilarity post-January 25 (higher Frobenius norms in Figure 3) correlates with changes in the geographic distribution of infections. (3) The temporal synchronization between network connectivity changes and regional infection curves validates the model's ability to capture mobility-dependent spread patterns. These relationships confirm that temporal variations in mobility networks directly influence epidemic trajectories.

Despite the improvements introduced by our model, some limitations must be acknowledged. First, we assume constant node populations during simulation, which is enforced through iterative flow adjustment. While this ensures numerical stability and population conservation, it abstracts away demographic changes or unrecorded migration events. Second, the model's outcomes are sensitive to the choice of parameters, particularly the transmission rate (β) and recovery rate (γ). These values, although commonly used in literature, may not reflect local epidemiological characteristics without proper calibration. Sensitivity analysis or data-driven parameter estimation (e.g., via optimization or machine learning) is essential for future work to validate and refine these assumptions.

4. Conclusions and future work

The primary goal of this study was to adapt the SIR metapopulation model for temporal mobility networks, incorporating dynamic inflow and outflow rates between nodes. To achieve this, we modified the classical SIR metapopulation equations to handle two temporal flow tensors—one for inflow and one for outflow—ensuring the conservation of the total number of hosts during simulations by adjusting rate matrices, whose sums must equal 1.

Simulation results using Baidu Mobility Data revealed key insights for epidemic control strategies in high-mobility contexts, such as during the Wuhan lockdown and the infection peak around the Lunar New Year. These findings suggest that interventions must

consider not only epidemiological parameters but also mobility patterns and sociocultural characteristics to optimize containment and prevention measures.

By incorporating dynamic inflow and outflow while ensuring host conservation, this adaptation offers a more flexible and precise approach to modeling disease spread in realistic mobility scenarios, expanding the applicability of the original model in epidemiology and mobility studies. We show that abrupt changes in mobility coincide with the observed responses in model dynamics.

This study opens avenues for future improvements, such as randomizing parameters β_i and γ, and incorporating social, demographic, and geographic features—like GDP, area, age, income, and occupation—to enhance model accuracy. These factors influence contact rates, susceptibility, and mobility, potentially revealing regional vulnerabilities and informing targeted interventions. Machine learning and optimization methods could also assist in parameter calibration and improve predictive performance.

Future work could also focus on replacing the simplified percentage-based approach with exact traveler counts for inflow and outflow, enabling flow rates to be calculated from real-world data. Additionally, the study could be expanded to analyze temporal networks at different time scales, comparing networks with varying temporal resolutions to understand the impact of granularity on disease spread. Metrics such as the Frobenius norm and Pearson correlation could quantify differences between simulations, offering a more detailed evaluation of model accuracy.

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